D-H - Algorithm



1. Z axis is achation axis
2. x_i is $\perp$ to z_i & z_{i-1} & intersects them
3. D_i is where x_i meets z_{i-1}
4. if revolute: $\theta = \theta_i$ & d is dist along z_{i-1} from O_{i-1} to O_i
5. if prismatic $d = d_i$ & θ is θ around z_{i-1} to align x_{i-1} to x_i
6. a_i is distance along x_i from b_i to O_i
7. α is rotation avond x_i to align z_{i-1} to z_i

	z_{i-1}		x_i	
i	θ_i	d_i	a_i	α_i

$$
{}^{i-1}T_i = \begin{bmatrix}
\cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\
\sin\theta_i & \cos\theta_i\cos\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\
0 & \sin\alpha_i & \cos\alpha_i & d_i \\
0 & 0 & 0 & 1
\end{bmatrix}
$$

Inverse matrix

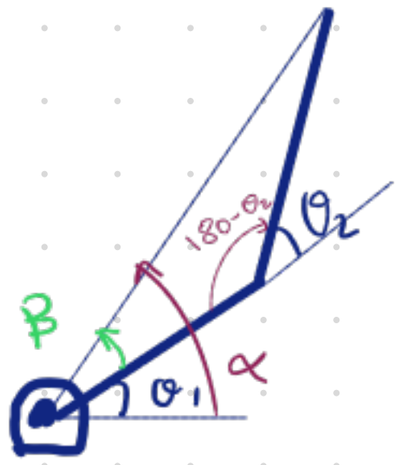
$$M = \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix}, \quad M^{-1} = \begin{bmatrix} R^T & -R^T T \\ 0 & 1 \end{bmatrix}$$

Inverse Kinematics

Given final transformation, get variables.

1- geometric method:

Apply Sin & cos rule on the triangle



2- Analytical method :

Ask Paul

Analytical method:

1) apply D-H algorithm ⇒ table

2) turn table into matricies ⇒ $M = {}^0A_1\ {}^1A_2\ {}^2A_3$ ...

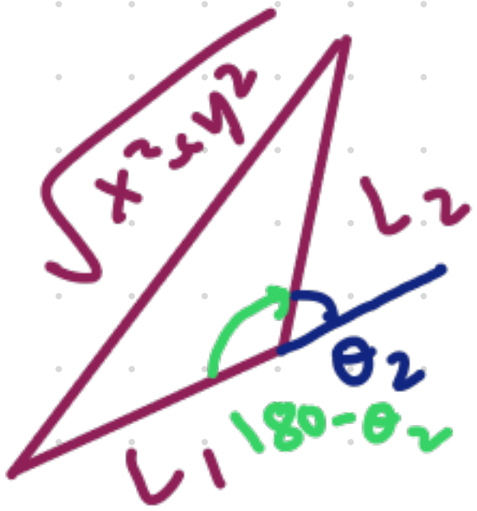
3) get variables one by one

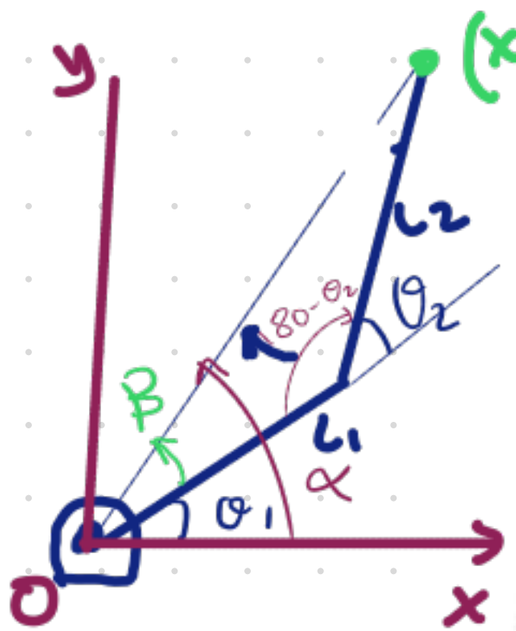
$({}^0A_1)^{-1} M = {}^1A_2\ {}^2A_3$ ... ⟶ Variable 1 ✓

$({}^1A_2)^{-1}({}^0A_1)^{-1} M = {}^2A_3$ ... ⟶ Variable 2 ✓

$({}^2A_3)^{-1}({}^1A_2)^{-1}({}^0A_1)^{-1} M =$ ... ⟶ Variable 3 ✓

$\sqrt{x^2+y^2}$





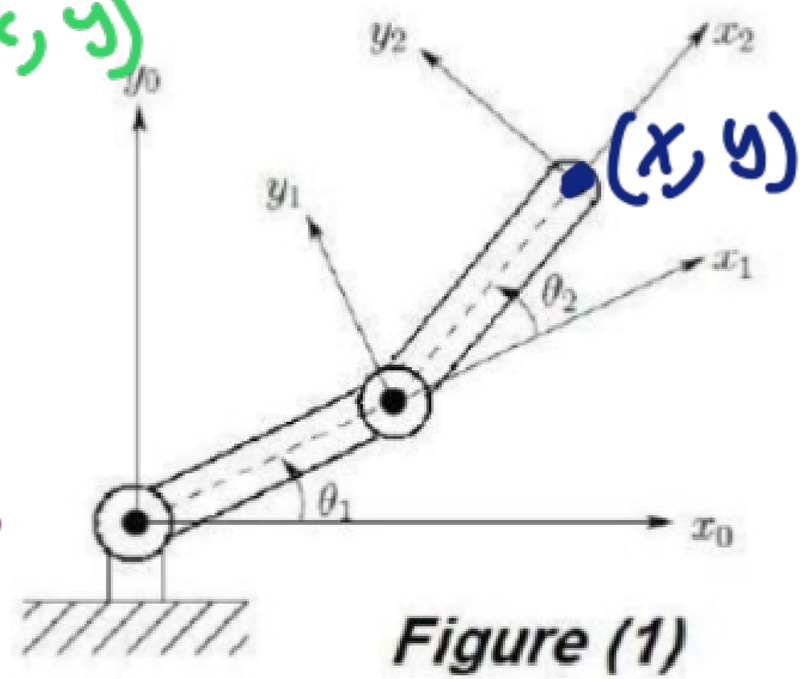


Figure (1)

$$\theta_2 = \cos^{-1}\left(\frac{L_1^2 + L_2^2 - (x^2+y^2)}{2L_1L_2}\right) \Rightarrow \theta_2$$

$$\alpha = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\beta = \alpha - \theta_1 = \tan^{-1}\left(\frac{y}{x}\right) - \theta_1 \;\; j \qquad \frac{\sin\beta}{L_2} = \frac{\sin(180-\theta_2)}{\sqrt{x^2+y^2}}$$

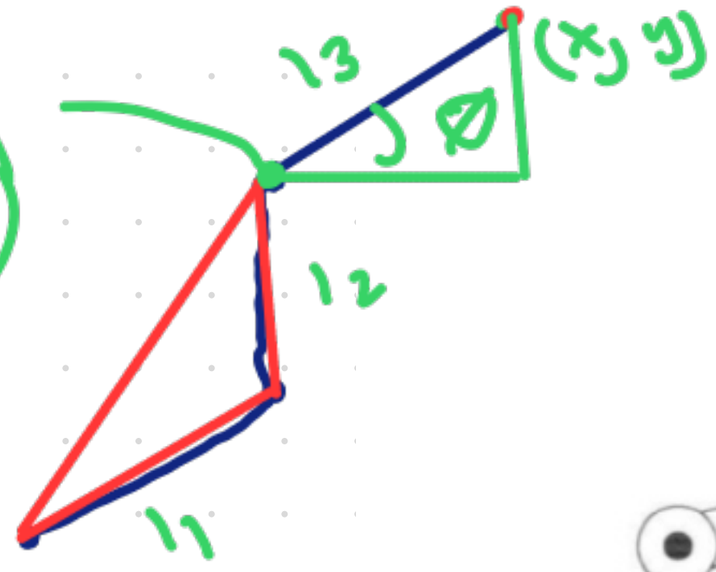
$\sin\theta_2$

$$\alpha - \theta_1 = \sin^{-1}\left(\frac{L_2}{\sqrt{x^2+y^2}}\sin\theta_2\right)$$

$$\theta_1 = \alpha - \sin^{-1}\left(\frac{L_2}{\sqrt{x^2+y^2}}\sin\theta_2\right)$$

$$\theta_1 = \tan^{-1}\left(\frac{y}{x}\right) - \sin^{-1}\left(\frac{L_2}{\sqrt{x^2+y^2}}\sin\theta_2\right)$$

$$\begin{pmatrix} x - l_3\cos\phi \\ y - l_3\sin\theta \end{pmatrix}$$



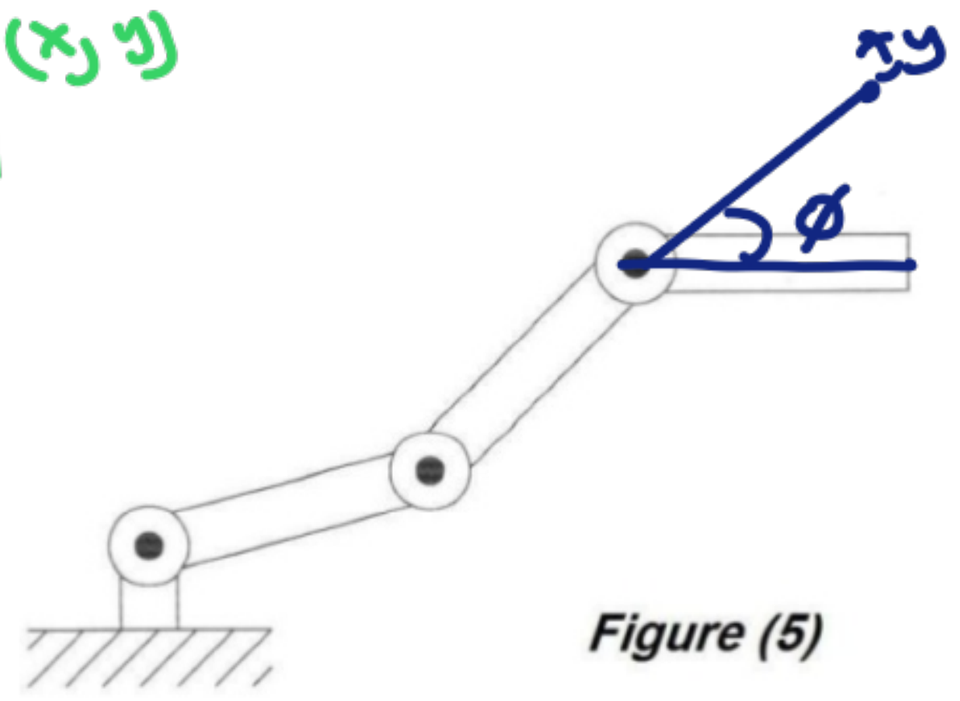
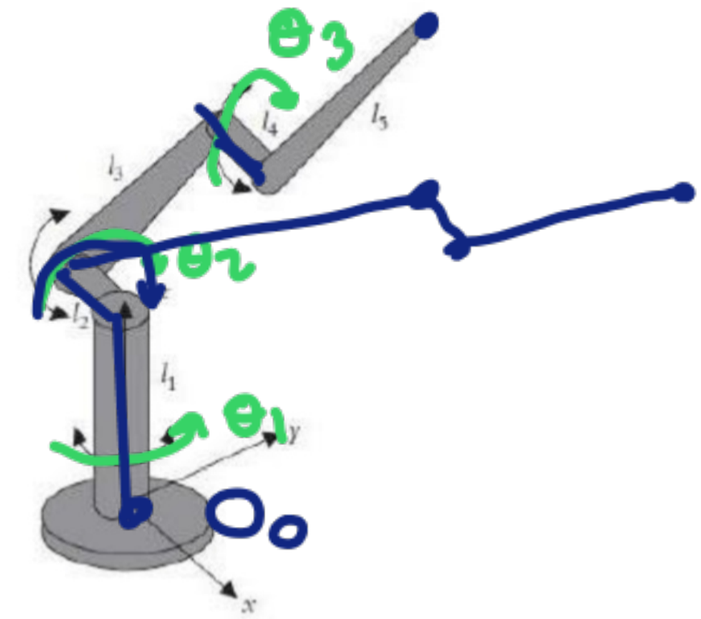
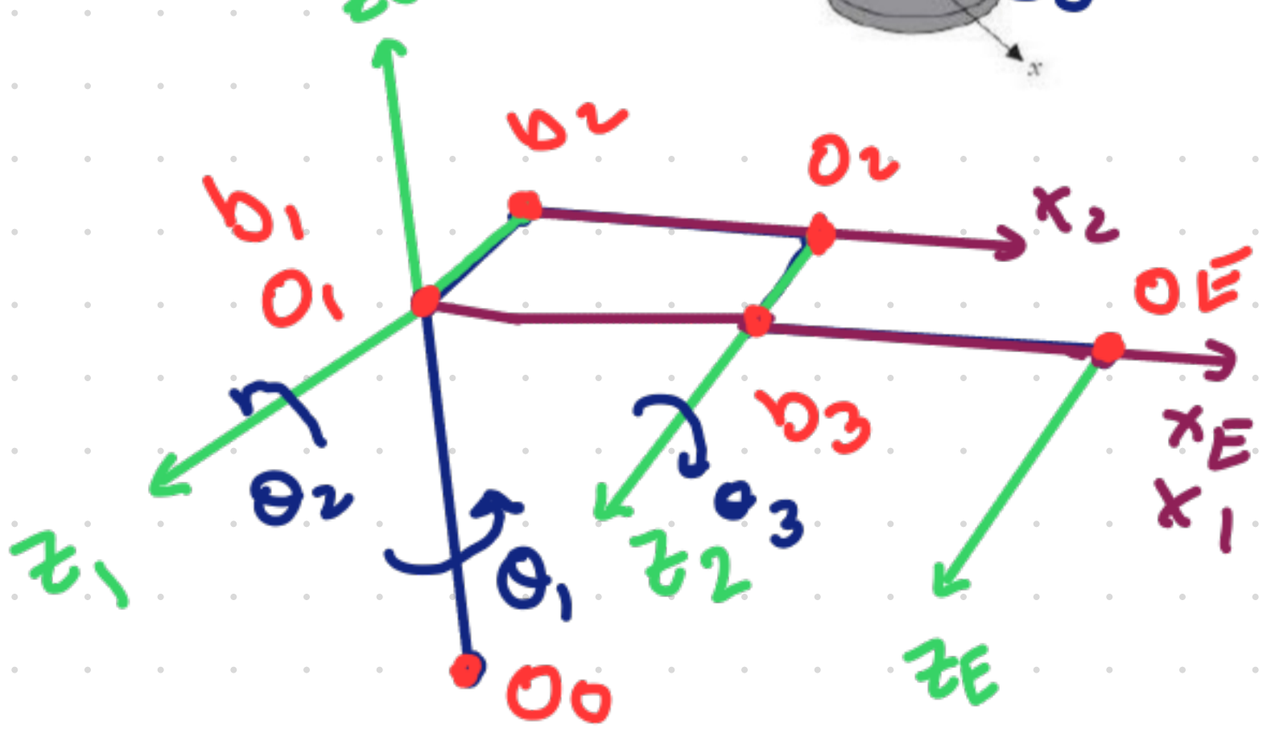


Figure (5)

$$T = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5. For the simple 3-DOF arm (l_1=8, l_2=2, l_3=9, l_4=2 and l_5=9) shown in Fig. 7, solve the inverse kinematic equations by **<u>using analytical method</u>** for the robot relative to frame 0 in order to calculate joint variable θ_1, θ_2 and θ_3 in terms of end-effector position p_x, p_y and p_z.





i	θ_i	d_i	a_i	α_i
1	θ_1	8	0	90
2	θ_2	-2	9	0
3	θ_3	2	9	0

$$
{}^{i-1}T_i = \begin{bmatrix}
\cos\theta_i & -\sin\theta_i \cos\alpha_i & \sin\theta_i \sin\alpha_i & a_i\cos\theta_i \\
\sin\theta_i & \cos\theta_i \cos\alpha_i & -\cos\theta_i \sin\alpha_i & a_i\sin\theta_i \\
0 & \sin\alpha_i & \cos\alpha_i & d_i \\
0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
{}^{0}T_{1} = \begin{bmatrix} \cos\theta_1 & 0 & \sin\theta_1 & 0 \\ \sin\theta_1 & 0 & -\cos\theta_1 & 0 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{1}T_{2} = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & 9\cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & 0 & 9\sin\theta_2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{2}T_{3} = \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & q\cos\theta_3 \\ \sin\theta_3 & \cos\theta_3 & 0 & q\sin\theta_3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{0}T_{E} = {}^{0}T_{1}\,{}^{1}T_{2}\,{}^{2}T_{E}
$$

$$
({}^{0}T_{1})^{-1} =
$$

$$({}^{0}T_{1})^{-1}\,({}^{0}T_{E}) = {}^{1}T_{2}\,{}^{2}T_{E}$$